Azure AI Foundry — Quick Reference

Overview

Azure AI Foundry is a platform for building, testing, deploying, and monitoring generative AI applications and agents. It brings together models, projects, tools, evaluations, and application workflows in one environment.

A typical Foundry workflow is: create a project → select or deploy a model → configure an agent or playground → add instructions and tools → test responses → evaluate and deploy.

Key Concepts

Projects

A project provides an isolated workspace for AI development. It can contain model deployments, agents, connections, evaluations, and other AI assets.

Models

Models provide capabilities such as text generation, reasoning, embeddings, and multimodal processing. A deployed model is made available to applications through an endpoint or project experience.

Agents

An AI agent combines a model with instructions, tools, and knowledge so it can perform tasks and respond using additional context.

Playgrounds

Playgrounds provide a browser-based interface for experimenting with prompts, system instructions, model settings, and responses.

Model Deployment

To use a model in an application, deploy an available model in the project. During deployment, the user typically selects the model, deployment name, region or capacity options, and other configuration supported by the environment.

Example deployment name: gpt-rag-model.

After deployment, verify that the deployment status is successful before selecting it in a playground or agent.

Prompting Basics

A system instruction defines the assistant's behavior, while a user message describes the task or question.

Example system instruction: You are an Azure AI assistant. Answer clearly, use only available knowledge when grounding is required, and state when information is unavailable.

Useful prompt techniques include giving context, defining the role, specifying the desired output format, providing examples, and stating constraints.

Responsible AI

AI applications should be designed to reduce inaccurate, unsafe, biased, or misleading outputs. Grounding responses with trusted knowledge sources can reduce hallucination risk but does not guarantee correctness.

For production systems, validate outputs, protect sensitive data, apply access controls, monitor usage, and test the application against expected and unexpected inputs.